

Affective Companion Robot with Speech-Driven Emotion Feedback

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Abstract
Smushie is a companion robot that adapts its non-verbal behaviour to users emotional states and preferences using a pre trained speech to emotion model and lightweight reinforcement style learning. Vocal input is mapped to discrete emotion categories, each linked to multiple vibrotactile and light patterns, while touch feedback acts as a binary reward that increases the selection weight of the active pattern. A small scale evaluation, combining classifier accuracy on held out speech samples with shifts in pattern selection probabilities from interaction logs, shows that the prototype can sense vocal affect, personalise its light–vibration responses over repeated use, and support a calming companion role while keeping all processing local and logged data minimal.

Introduction
Social robots are expected to interpret and respond to human emotions in order to support engaging, supportive interactions rather than purely functional exchanges. Work in affective computing shows that speech is a rich source of affective cues and that soft computing or machine learning methods can infer users emotional states from vocal features. Human robot interaction studies highlight the importance of clear, non verbal behaviours such as light patterns and haptic feedback for communicating a robot’s internal state and fostering a sense of empathy.

This project investigates a compact companion robot that combines speech-based emotion recognition with adaptive vibrotactile and light responses. A simple learning mechanism adjusts the frequency of expressive patterns according to touch feedback, tailoring behaviour to individual preferences while keeping the underlying emotion model fixed and exploring whether such low-complexity, personalised non-verbal expressions can improve user engagement, comfort, and perceived emotional attachment in everyday interaction.

- Algorithm Design**
- Python records short audio clips, extracts chroma/mel features, and uses a Kaggle RAVDESS trained classifier to predict an emotion label (happy, calm, sad, angry, or neutral).
 - For that emotion, a policy module selects one of two light–vibration patterns and sends a compact code (emotion letter plus pattern index) over serial to the Pico, which plays the pattern on the cube.
 - After each interaction, a lightweight reinforcement update boosts the chosen pattern’s weight when touch is detected and slightly decays it otherwise, then clamps and renormalises weights so the policy adapts while the classifier stays fixed.

Pseudocode

```
1 open serial connection to Pico
2 load pre-trained speech-emotion classifier
3 initialise pattern_weights[emotion][pattern_id] = 1.0
4 while system is running
5   record fixed-length audio segment
6   extract features and predict emotion
7   choose pattern =
   sample_pattern(pattern_weights[emotion])
8   send compact code (emotion_letter, pattern_index) to Pico
9   Pico plays corresponding vibration + LED pattern
10  wait feedback window
11  read touch feedback (present / absent)
12  if feedback is present then
13    | increase pattern_weights[emotion][pattern]
14  else
15    | apply slight decay to pattern_weights[emotion][pattern]
      | (down to minimum)
16  end
17  renormalise pattern_weights[emotion]
18 end
```

Literature Review
Affective computing treats emotion as an information signal that can be sensed and modelled using soft computing methods applied to modalities such as speech prosody. Social robotics work argues that robots should adapt to users preferences and affective states over time to sustain engagement and rapport, often through simple, legible behaviours rather than complex humanoid embodiment. Studies of non-verbal HRI demonstrate that vibrotactile feedback and coloured light patterns can communicate affect and comfort, especially when combined with friendly form factors and soft materials [1].

Interaction Design & Usability
The artefact is a palm-sized, slightly weighted cube that can be comfortably grasped, then covered with a soft fuzzy sleeve to provide pleasant haptic texture and diffuse vibrotactile feedback, supporting approachable, low-threat interaction similar to a small pet or comfort object [2].

The system runs continuously in the background, monitoring ambient speech and activating coloured light and vibration patterns only when the speech emotion module detects angry, happy, or sad vocal expressions, so the cube gently attracts attention and invites handling without requiring explicit commands. Users can simply pick up, hold, or absent mindedly pat the cube.

These gentle interactions are intended to feel soothing and supportive, echoing findings that touching soft social robots can reduce pain and stress [3]. Binary indications of engagement that drive the reinforcement style adaptation of future patterns to

customize to the users preference.Different light–vibration patterns encode the inferred emotion and invite the user to interact, consistent with work on ‘tactile emoticons’ that convey social emotions through patterned touch feedback [4]

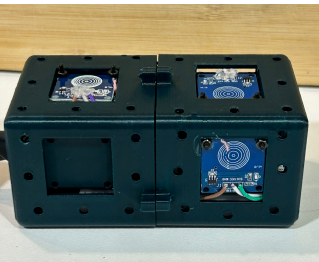


Figure 1: Smushie Interior

Learning and Adaptation Mechanism
For each emotion category, the system maintains a probability distribution over available patterns, initialised uniformly and updated after every interaction. Touch feedback (contact detected on the capacitive pads during the response window) is treated as a binary reward signal (present vs absent) that increments or slightly decays the weight of the selected pattern before renormalisation. Weights are clamped between minimum and maximum values prior to renormalisation to prevent degenerate policies where a single pattern suppresses all others, yielding a compact policy level reinforcement mechanism that adapts interaction parameters in response to user behaviour.

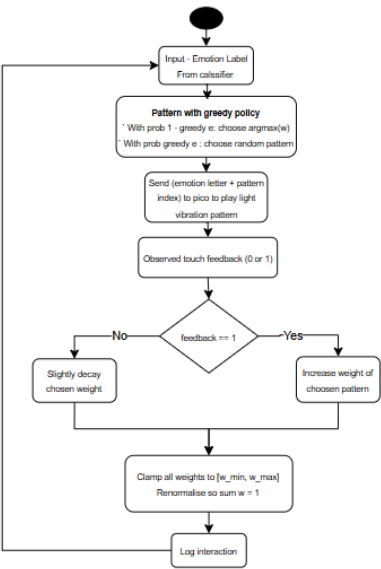


Figure 2: Activity diagram of reinforcement logic for pattern selection

Data Analysis
The system evaluates affect sensing using the classifier’s predicted labels on a heldout subset of RAVDESS samples and reports indicative accuracy for the target emotion classes. Interaction logs record pattern selection probabilities per

emotion and the frequency of touch versus no touch events, making it possible to visualise how the policy shifts toward user preferred light vibration patterns (for example, between happy_0 and happy_1). Informal pilot observations, such as voluntary re-engagement with the artefact.Vibration patterns are designed to be gentle and rhythmic, aiming to calm rather than startle, similar to work on vibrotactile ‘heartbeat’ cues for socially assistive robots.Interaction logs are exported to CSV and a Python script plots bar charts of the current pattern-selection probabilities per emotion, visually highlighting which light–vibration patterns have become favoured over time through user feedback [5]

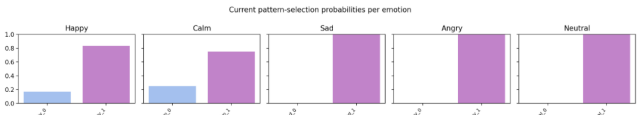


Figure 3: Graph of Learned pattern selection per emotion

Cybersecurity and Data Handling
Raw audio from the microphone is processed locally and is not stored, only abstracted interaction logs (interaction ID, emotion–pattern label, touch feedback, and current pattern weights) are stored on the machine, without personal identifiers.Communication between Python and the Pico uses a simple, documented serial protocol, and access to the logging directory is restricted to authorised users of the development machine to reduce the risk.

- Limitations**
- Requires a host computer for speech processing.
 - Limited set of light–vibration patterns and rigid cube form.
 - Only short, informal user testing so far.

- Future Work**
- Move the full pipeline onto an ESP32 or Raspberry Pi.
 - Add richer patterns and softer, squeezable enclosures.
 - Run longer-term studies of use in everyday settings.

Bibliography

- [1] I. S. Koh and H. S. Kang, “Effects of Intervention Using PARO on the Cognition, Emotion, Problem Behavior, and Social Interaction of Elderly People with Dementia,” Journal of Korean Academy of Community Health Nursing, vol. 29, no. 3, p. 300, 2018, doi: 10.12799/jkachn.2018.29.3.300.
- [2] P. Ihmäki and K. Heljakka, “Robot Pets as “Serious Toys”: Activating Social and Emotional Experiences of Elderly People,” Information Systems Frontiers, pp. 1–15, Aug. 2021, doi: 10.1007/s10796-021-10175-z.
- [3] N. Geva, N. Hermoni, and S. Levy-Tzedek, “Interaction Matters: The Effect of Touching the Social Robot PARO on Pain and Stress is Stronger When Turned ON vs. OFF,” Frontiers in Robotics and AI, vol. 9, July 2022, doi: 10.3389/frobt.2022.926185.
- [4] A. Saramandi et al., “Tactile emoticons: Conveying social emotions and intentions with manual and robotic tactile feedback during social media communications,” PLOS ONE, vol. 19, no. 6, p. e304417, June 2024, doi: 10.1371/journal.pone.0304417.
- [5] “Soothing Sensations: Enhancing Interactions with a Socially Assistive Robot through Vibrotactile Heartbeats Research supported by the UKRI Centre for Doctoral Training in Socially Intelligent Artificial Agents, Grant Number EP/S02266X/1.” [Online]. Available: <https://arxiv.org/html/2410.07892v1>